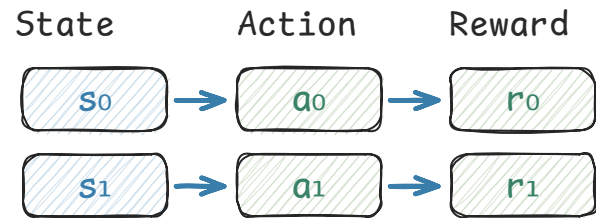
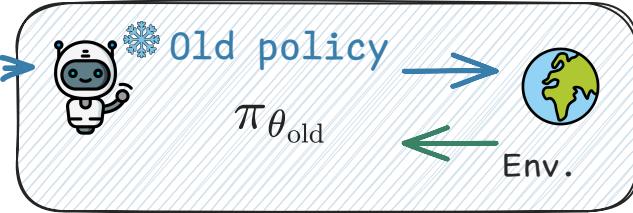


PPO Proximal policy optimization · PPO-Clip

Collect fresh rollouts



Store data + old log-probs

Estimate advantages



Learned value baseline

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$
$$\hat{A}_t = \sum_{l=0}^{T-1-t} (\gamma\lambda)^l \delta_{t+l}$$

GAE combines discounted TD residuals

Advantage + value targets

Hold estimates fixed during the epochs.

Optimize actor + critic



Trainable actor + critic

K epochs · shuffled mini-batches



Keep the rollout-policy denominator fixed

$$\rho_t = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$$

First-order optimizer (e.g., Adam)

Refresh the old policy, then collect new data.

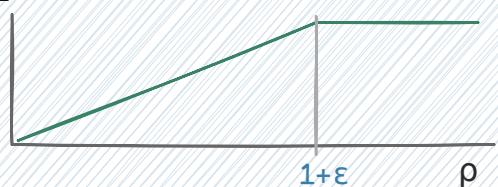


Clipped surrogate: a conservative incentive for policy updates

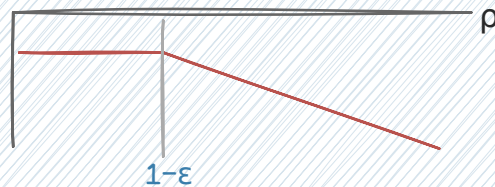
$$L^{\text{CLIP}} = \widehat{\mathbb{E}}_t \left[\min \left(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$



$A > 0$: clip high ratios



$A < 0$: clip low ratios



Value fitting + entropy bonus

$$J = L^{\text{CLIP}} - c_V \widehat{\mathbb{E}}_t \left[(V_{\phi}(s_t) - \hat{R}_t)^2 \right] + c_H \widehat{\mathbb{E}}_t [H(\pi_{\theta}(\cdot | s_t))]$$

Fit return targets; optionally encourage exploration.